

Profiling flavor characteristics of cold brew coffee with GC-MS, electronic nose and tongue: effect of roasting degrees and freeze-drying

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Abstract

BACKGROUND: Roasting is an important process in the formation of coffee flavor characteristics, which determines the quality of coffee and consumer acceptance. However, the influence of roasting degree on the flavor characteristics of cold brew coffee has not been fully described.

RESULTS: In the present study, the flavor characteristics of cold brew coffee with different roasting degrees were compared in detail by using chromatographic and electronic sensory approaches, and the flavor changes induced by freeze-drying were investigated. Pyrazine and heterocyclic compounds were the main aroma compounds in coffee, and gradually dominated with the increase of roasting. Pyridine was consistently present in cold brew coffees of different roasting degrees and showed significant gradient of quantity accumulation. Aroma compounds such as pyrazine, linalool and furfuryl acetate were the main contributors to coffee roasting, floral and fruity flavor. Freeze-drying preserved the fruity and floral aromas of medium-roasted cold brew coffee, whereas reducing the bitterness, astringency and acidity properties that are off-putting to consumers.

CONCLUSION: The higher consumer acceptance and enjoyment in medium roast cold brew coffee may be related to its stronger floral and fruity aroma. The aroma profile qualities of freeze-drying processed medium roasted cold brewed coffee were more dominant and more suitable for freeze-drying processing than medium dark roasting. Application of freeze-drying for cold brew coffee will promote the convenience of drinking. The present study provides valuable technical guidance in improving the flavor and quality of cold brew coffee, and also promotes its commercialization process.

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Supporting information may be found in the online version of this article.

Keywords: cold brew coffee; roasting degree; freeze-drying; volatile compounds; OPLS-DA

INTRODUCTION

Coffee is one of the top three most-consumed beverages in the world, which is popular for its unique flavor and functional characteristics. Hot brewing has long been the conventional way of drinking coffee globally, regardless of people having different cultural and social backgrounds, as well as personal preferences.¹ However, cold brewing has become increasingly popular among consumers. Unlike hot brewing, cold brew coffee requires water no higher than room temperature and takes a long time to brew, ranging from 8 to 24 h.² This special process contributes a less bitter and a less acidic flavor to cold brew coffee, which is more in line with the taste needs of the general public, driving its latest consumption craze.

With the great popularity of cold brew coffee, the method of obtaining better coffee flavor is the pursuit of many processors. In general, the flavor of coffee depends on factors such as bean cultivars, quality, post-harvest processing, roast type and storage.³ Roasting is an essential and vital part of coffee processing. It should be noted that raw coffee beans do not have aroma by themselves and need to be roasted and processed to undergo a

series of chemical reactions such as the Maillard reaction and the caramelization reaction to give the coffee its characteristic color, flavor and mouthfeel.⁴ In general, an increase in the degree

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of roasting results in a significant increase in the type and content of aroma compounds, especially the furans, pyrazines and pyrroles that contribute to the roasted, nutty and burnt aromas of coffee.⁵ However, furfural and derivatives are found at a relatively higher level in mild-roasted coffee, but with a decrease in their higher-roasted counterparts. A poor Maillard reaction will also lead to insufficient production of aroma compounds, especially volatile organic compounds with the characteristics of nuts, coconut and chocolate, thus reducing the aroma quality of coffee.⁶ Lightly roasted coffees tend to have a predominantly sour taste, with bitter, burnt and acrid notes gradually dominating the sensory panel as the roast level increases.⁷ It should be noted that the above findings are mainly derived from studies on hot brew coffee, and studies on the effect of roasting degree on the flavor profile of cold brew coffee are limited. Although the difference between hot and cold brew coffee is mainly in the temperature of the water, the impact of the difference in water temperature is huge. On the one hand, high temperatures increase the kinetic energy of water molecules and promote the leaching of compounds from the coffee powder. On the other hand, high temperatures also promote the differences in the release and volatilization of volatile compounds, and result in significant sensory perception differences.⁸ For example, a study found that cold brew coffee with immersion extraction had a higher abundance in total furans and pyrazines than hot brew coffee by the French press method.⁹ By applying gas chromatography-mass spectrometry and odor activity value calculation, it was found that most pyrazines contributed more to the aroma profile of cold brew coffee than hot brew coffee.¹⁰ This makes it clear that, although some detailed data and findings on the effect of roast degree for coffee flavor profile have been obtained in hot brew coffee, these findings do not fully reflect the effect of roast degree on the flavor profile of cold brew coffee. Therefore, elucidating the differences in flavor profiles of cold brew coffee with different roasting degrees is also a topic worth exploring. Seninde *et al.*¹¹ and Maksimowski *et al.*¹² have compared the differences in quality and organoleptic characteristics of cold brew coffees with different roasting degrees, but medium and dark roasted coffees were used as the main subjects in their studies. When limited to a narrow range of roasting degrees, it is only through simple comparisons that aroma compounds with differences can be found, and it will be difficult to truly identify the key aroma components that change significantly with the degree of roasting. The elucidation of the key aroma components that determine the flavor qualities or differences in cold brew coffee at different roasting degrees is important because it can guide coffee producers to better control the consistency of product quality. Moreover, it is important for optimizing the coffee roasting process. This is because, even at the same degree of roasting, different roasting profiles (time–temperature) can result in very different volatile and non-volatile compounds in the coffee.⁹ When the key aroma compounds that determine or characterize the degree of coffee roasting are clarified, this will provide more scientific guidance for the improvement of the roasting process and roasting conditions. Therefore, a more detailed delineation of different roasting degrees and exploration of the differences in key aroma compounds at different stages will lay a solid material foundation for elucidating the quality differences in the flavor of cold brew coffee at different roasting degrees. More intriguingly, according to an online survey, more than 50% of respondents indicated a preference for medium-roasted coffee beans for cold brew coffee making.² The medium roasted coffee beans

can become the products favored by consumers, which must be closely related to the flavor compounds involved. However, knowledge of the compounds responsible for the differences in the sensory characteristics of cold brew coffee at different roasting degrees is very limited, making it difficult to truly reveal the material basis of the flavor components that determine consumer acceptance and consumption preferences. This is not conducive for coffee producers to improve the quality of coffee and further develop products that cater to consumers' drinking preferences. Therefore, a comprehensive analysis of cold brew coffee of different roasting degrees is needed.

Cold brew coffee is usually consumed immediately after preparation, or after a short storage period under refrigerated conditions. Making cold brew coffee privately at home is currently the main consumption mode of consumers, and there is still a huge supply gap for the industrial production of cold brew coffee.² Although the process of making cold brew coffee is not cumbersome, it is time consuming (8–24 h, as mentioned above) and the convenience of consumption is the biggest pain point for consumers. In addition, Kwok *et al.*¹³ also expressed concern about the practical aspects related to hygiene and food safety of the cold brew coffee. To promote the commercialization of cold brew coffee, Bellumori *et al.*¹⁴ compared the effects of different stabilization techniques on the flavor and microbiological safety of cold brew coffee and found that UV irradiation maximized the preservation of the chemical and sensory characteristics of cold brew coffee, although safety was difficult to ensure. The other strategy is to dry the cold brew coffee and process it into powder that consumers can re-dissolve with water when drinking. The choosing of thermal drying method appears to seriously alter the flavor quality of cold brew coffee. This is because a study by Liu *et al.*¹⁵ found that only the non-thermal treatments of membrane filtration and ultra-high-pressure were used to better maintain the sensory quality of cold brew coffee compared to pasteurization and high-temperature instantaneous sterilization. At the same time, Liu *et al.*¹⁵ also pointed out that when combining cold brew coffee with thermal processing treatments, this goes against the cognitive positioning of consumers towards cold brew coffee products and may reduce consumer acceptance. Furthermore, freeze-drying can yield high quality products and maintain nutrients, color and flavor, and has been used more often in fruits and vegetables processing.¹⁶ Compared with thermally dried products, freeze-drying can greatly maintain the original nutrients and volatile components and obtain a higher quality coffee product as a result of the low temperature and oxygen-free operating conditions.¹⁷ Various ways and methods have been explored to improve the efficiency of freeze-drying coffee,^{18,19} but the suitability of freeze-drying for cold brew coffee is unknown, and there is no direct reference for the effect on flavor characteristics in particular.

In the present study, headspace solid-phase microextraction was used to comparatively analyze the aroma characteristics of cold-brewed coffee with different roasting degrees. Based on this, key differential aroma compounds were identified and the material basis and aroma characteristics that determine the high acceptability of medium roast cold brew coffees were elucidated. The differences in aroma and taste characteristics of cold brew coffee before and after freeze-drying were also investigated to clarify the processing applicability of freeze-drying technology, with a view to providing guidance for the production and processing of high-quality cold brew coffee and promoting its commercialization.

MATERIALS AND METHODS

Materials

The same commercial batch of Arabica coffee beans (Pacific, Shanghai, China) blended from Costa Rica, Colombia and Indonesia were selected for coffee brewing. The roasted coffee beans were processed by Jiangsu Yuanli Jihua Beverage Co., Ltd (Jiangsu, China) and cold brew coffee samples were prepared for experimental analysis. Fresh coffee beans were roasted at 230 °C and the time was controlled to achieve different levels of roast. The color change was determined using an NR110+ (Sanenshi Technology Co., Ltd, Guangzhou, China) colorimeter, and the roasting degree was determined based on the results of the measurements to confirm that the coffee beans were in the desired state of roasting. According to the company's roasting standards, the roasting degree of cold brew coffee was divided into light roast (LR), medium-light roast (MLR), medium roast (MR), medium-dark roast (MDR), dark roast (DR) and ultra-dark roast (UDR). The grinding of coffee beans and the preparation of cold brew coffee refer to the research conducted by Cai *et al.*¹⁰ In short, coffee beans were grounded by the semi-automatic grinder (E10; HERO, Beijing, China) and sieved to obtain 30–60 mesh coffee powders for cold brew coffee. Coffee powders were placed into a 300-mesh filter bag and let soak in purified drinkable water at room temperature (20 ± 3 °C) for 8 h.

Freeze-drying processing of cold brew coffee

The cold brew coffee stock was frozen at -80 °C for 12 h and then the frozen coffee was placed into a vacuum freeze-dryer (Four-Ring Science Instrument Plant Beijing Co., Ltd, Beijing, China) at -40 °C for 48 h. The freeze-dried coffee was milled and transferred to bags. Cold brew coffee powder was diluted with pure drinking water to standardize to 2% total dissolved solids for sequential instrumental analysis.¹⁰

Extraction of volatile compounds

The extraction of volatile aroma compounds from cold brew coffee was performed by headspace-solid phase microextraction (HS-SPME) as described by Cai *et al.*,¹⁰ with minor modifications. Cold brew coffee (5 mL) was transferred to a 20-mL headspace bottle containing 2 g of NaCl and 100 μ L of 3-octanone (8.2 mg L^{-1} , as an internal standard). The bottles were sealed with polytetrafluoroethylene silicone septa and equilibrated with stirring at 40 °C for 10 min. Next, 50/30 μ m divinylbenzene/carboxy/polydimethylsiloxane SPME fiber was exposed to the coffee headspace for 20 min at the same temperature. After 20 min of extraction at 40 °C, the fiber was immediately desorbed into the gas chromatography injector port at 250 °C for 5 min.

Gas chromatography-mass spectrometry (GC-MS)

GC-MS analysis was carried out using an Agilent 7890 gas chromatography system (Agilent Technologies, Santa Clara, CA, USA) equipped with an Agilent 5975C series mass spectrometer. The DB-WAX fused silica capillary columns (30 m $\times$ 0.25 mm $\times$ 0.25 μ m; Agilent Technologies) was programmed from 60 °C (withholding for 5 min) to 180 °C at 3 °C min^{-1} and then rose to 250 °C at 10 °C min^{-1} (withholding for 5 min). The carrier gas was helium at a rate of 1 mL min^{-1} constant flow. For mass spectra analysis, it was operating in electron impact mode of 70 eV with a scan range of m/z 55–550. Identification of volatile compounds was achieved by matching the mass spectra of the respective compounds to the NIST12 database.

The quantification of volatile compounds was performed using peak areas normalized with 3-octanone added to each sample as an internal standard.

Identification of aroma-active compounds

Odor activity value (OAV) was the ratio of concentration to odor threshold. In most cases, an OAV value >1 means the compound contributes to the overall coffee aroma profile.

Electronic tongue analysis

Electronic tongue analysis was performed using an SA402B taste sensing system (Intelligent Sensor Technology, Inc., Kanagawa, Japan) equipped with an autosampler, a reference electrode and a multichannel lipid/polymer membrane electrode. Cold brew coffee sample (60 mL) was placed in the automatic sampling analysis device. The potential differences between sensors coated with each individual compound and the Ag/AgCl reference electrode were measured in equilibrium. Each collection time was 120 s and data were collected once every 1 s.

Electronic nose analysis

The odor of coffee samples was measured using the electronic nose system PEN-3 (Airsense, Schwerin, Germany), which consists of a sample collection unit and a metal oxide semiconductor sensor. A sample (1 mL) was transferred to a measuring vial and left to stand for 30 min at room temperature, and the top gas was aspirated for detection using the automatic inlet system of the electronic nose. The performance description of each sensor is shown in the Supporting information (Table S1).

Statistical analysis

All analyses were conducted in triplicate. The relevant results were denoted as the mean $\pm$ SD. Significant differences among groups were performed statistically by one-way analysis of variance using the SPSS, version 23.0 (IBM Corp., Armonk, NY, USA). The orthogonal partial least-squares discriminant analysis (OPLS-DA) was performed using SIMCA, version 14.1 (Sartorius, Umea, Sweden). The R software package (R Foundation, Vienna, Austria) was used to conduct the heatmap analysis and cluster analysis on the basis of signal abundances in cold brew coffee.

RESULTS AND DISCUSSION

Analysis of aroma composition of cold brew coffee

To investigate the aroma characteristics of cold brew coffee with different roasting degrees, the aroma compound composition and relative content of six coffee samples were analyzed and identified using headspace solid-phase microextraction coupled with GC-MS. As shown in the Supporting information (Table S2), 63 aroma compounds were detected, including seven aldehydes, four alcohols, three ketones, three esters, 10 phenols, 15 pyrazines, 18 heterocyclic compounds, one hydrocarbon and two ethers.

From Fig. 1, the total content of aroma compounds showed an overall increasing trend with the deepening of roasting, reflecting the role of the roasting process in promoting coffee aroma production.²⁰ However, the total amount of aroma compounds was slightly reduced in the medium-dark roasted coffee compared to the medium roast. In this regard, the increase in the degree of roasting (from medium to medium-dark) did not reflect the promotion on the formation of aroma compounds. During the high temperature roasting process, in addition to the complex chemical reactions that occur to produce various types of aroma

compounds, the shape of the coffee beans also undergoes significant changes, with the beans swelling and creating numerous cracks on the surface.²¹ The cracking of the beans will cause the escape of aroma compounds. The increase from medium to medium-dark roasting may have a very limited effect in promoting the production of aroma compounds, but the aroma escape as a result of bean cracking is significant,³ and this may be an important reason for the slight decrease in aroma compound content. For dark and ultra-dark roasted coffee beans, the ultra-high intensity high-temperature roasting process led to the intense generation of more aroma compounds. Although it is difficult to avoid the dissipation and loss of some aroma compounds, the increase effect of high content of total aroma compounds was much higher than the loss effect caused by dissipation, which made the overall increase trend of aroma compound content still observable in dark and ultra-dark roasted coffee beans. With the increase of roasting degree, the content of pyrazines and heterocyclic compounds increased significantly and gradually dominated. This is because of the Maillard, Strecker, pyrolysis and other chemical reactions in the roasting process, resulting in a large number of new compounds, including furans, pyridines, indoles and other compounds.²⁰

Comparison of aroma composition of cold brew coffee with different roasting degrees

With aroma compounds as the dependent variable and roasting degree as the independent variable, OPLS-DA analysis allowed effective differentiation of cold brew coffee samples with different roasting degrees. The fit index of the independent variable (R^2_X) was 0.953, the fit index of the dependent variable (R^2_Y) was 0.943 and the model prediction index (Q^2) was 0.811, with R^2 and Q^2 exceeding 0.5, indicating acceptable model fit results.²² After 200 permutation tests, the intersection of the Q^2 regression line with the vertical axis was less than zero, indicating that there was no overfitting of the model and the model validation was effective, and the results were considered acceptable for the analysis of roast degree identification of cold brew coffee aroma. As can be seen from the Fig. 2, the different cold brew coffee groups are ranked in order from right to left as the roast degree increases, indicating that there are significant differences in the composition of aroma compounds among cold brew coffees with different roasting degrees. It is worth noting that medium and medium-depth roasted coffee are closely arranged, and there is strong similarity in the composition of aroma compounds between them. This is consistent with the results in Fig. 1, further indicating that the difference between them may be mainly a result of the difference in content caused by the aroma release.

Because of the distinction between cold brew coffees with different roasting degrees, the differential aroma compounds were screened according to the criteria of $P < 0.05$ and variable importance in projection > 1 to clarify the material basis of the differences in aroma characteristics of cold brew coffees (Fig. 3). From the screened differential aroma compounds, it can be seen that fewer aroma compounds were present in the lightly roasted samples, which is consistent with the results of Fig. 1. Although the content of undecane in lightly roasted coffee is significantly different from the other groups, alkanes do not usually have aromatic characteristics and their contribution to the overall aroma profile of coffee is negligible.²³ Methyl-pyrazine, 1-(1-cyclohexen-1-yl)-ethanone, 5-methyl-2-furancarboxaldehyde and 2,3-dihydrobenzofuran gradually increased in the coffee samples with increasing roasting degrees. However, excessive roasting or

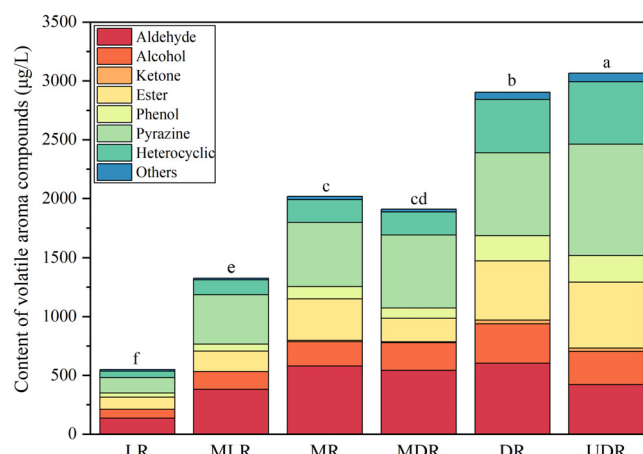


Figure 1. Composition and content of aroma compounds in cold brew coffee at different roasting degrees. Different lowercase letters demonstrate a significant difference at different roasting degrees ($P < 0.05$). LR, light roast; MLR, medium-light roast; MR, medium roast; MDR, medium-dark roast; DR, dark roast; UDR, ultra-dark roast).

prolonged exposure will inevitably lead to the loss of aroma compounds,²⁴ such as 5-methyl-2-furancarboxaldehyde and 1-(1-cyclohexen-1-yl)-ethanone. Interestingly, pyridine was consistently present in various coffee samples and was highest in ultra-deep roasted samples, with potential as a marker indicating the degree of roasting. It was also found that the medium roast samples possessed the richest variety of aroma compounds, especially containing linalool, 2-ethyl-3,5-dimethylpyrazine, benzeneacetaldehyde and 3-thiophenecarboxaldehyde, which are significantly different from other coffee samples. They are typical aromatic compounds formed by Maillard reaction and the Strecker degradation pathway,²⁵ presenting pleasant fragrance such as floral fragrance and burnt sweet fragrance. Among them, 2-ethyl-3,5-dimethylpyrazine has been confirmed as the potent odorants in the final products.²⁶ The advantages of medium roast cold brew coffee in the type and content of aroma compounds may be the important reason why its flavor characteristics are different from other roasting degrees. This needs to be further analyzed through the flavor characteristics of aroma compounds and elucidate the reasons for their high popularity in consumption.

The contribution of aroma compounds to the overall aroma of coffee depends on their concentration and odor threshold, and the level of aroma compounds alone is not a sufficient basis for determining the aroma characteristics of cold brew coffee. OAV is a reasonable tool for aroma potency assessment based on the equilibrium between air and the food matrix.²³ Table 1 presents the 12 volatile compounds with OAV ≥ 1 and their reliable odor thresholds reported before, indicating they make powerful contributions to the overall aroma. Pyrazines occupy the majority of these compounds since many of them possess a low odor threshold. Previous studies have shown that pyrazine is the major contributor to the aroma of hot brew coffee³ and, from this study, the same holds true for cold brewed coffee. In the light roast samples, only 4-vinyl-2-methoxyphenol and 3-ethyl-2,5-dimethylpyrazine contributed weakly to the aroma quality of cold brew coffee. As the roast degree increases, the OAV of pyrazines (except 2-ethyl-3,5-dimethylpyrazine) gradually increases, which will give the cold brew coffee a more intense nutty and baking flavor.²⁷ Previous studies have shown that some pyrazines

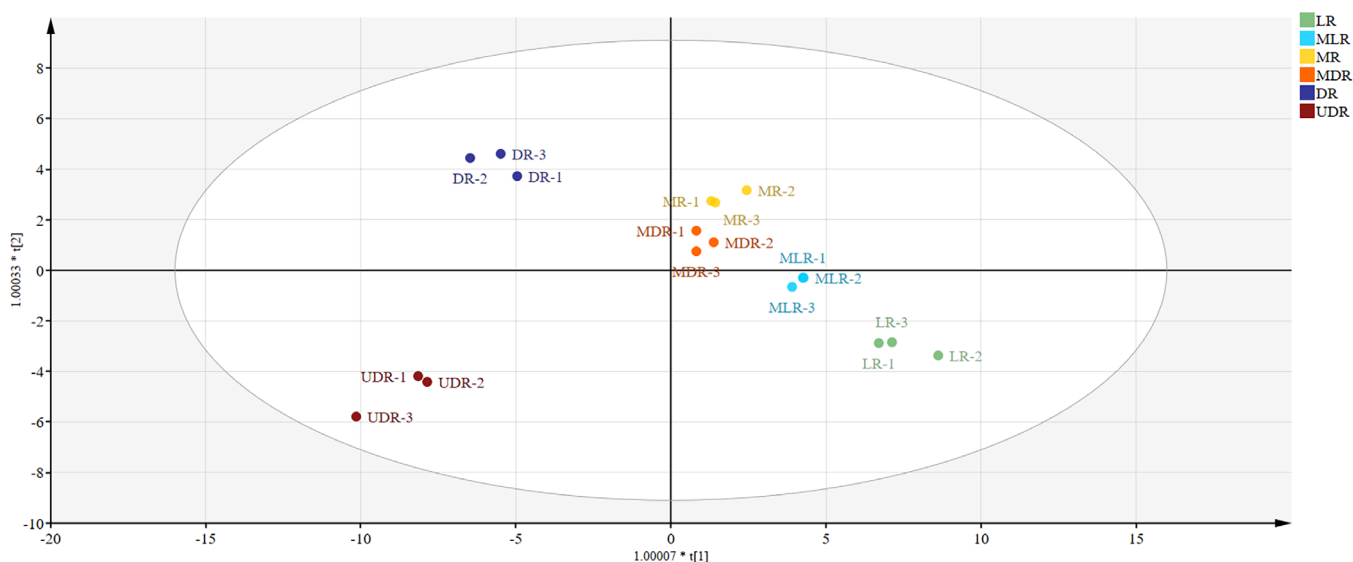


Figure 2. Results of orthogonal partial least squares discriminant analysis (OPLS-DA) of volatile compounds in cold brew coffee of different roasting degrees. LR, light roast; MLR, medium-light roast; MR, medium roast; MDR, medium-dark roast; DR, dark roast; UDR, ultra-dark roast.

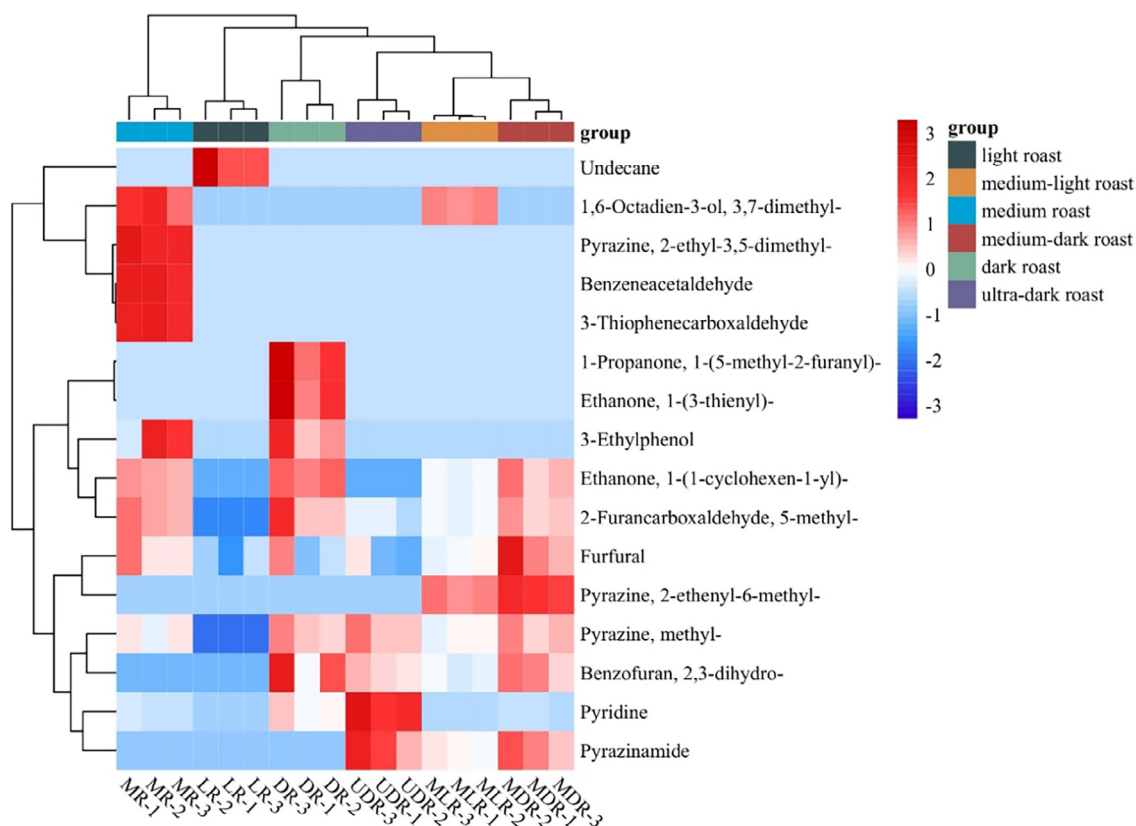


Figure 3. Heat map of the types and contents of differential aroma compounds in cold brew coffees of different roasting degrees.

exhibit a rapid increase at the beginning of the roasting process, while after excessive roasting, pyrazines show a significant decrease in content as a result of degradation and release.^{28,29} Although the present study also found that 2-ethyl-3,5-dimethylpyrazine had the highest OAV value in moderately roasted coffee, it was distinguished from other studies in that this compound was not detected in other groups. This may be

because of further amplification of the difference caused by the cold brew method. Overall, medium roast cold brew coffees have the highest OAV, with particularly pronounced nutty and roasted flavors. In addition, one of the important reasons why the flavor characteristics of cold brew coffee differs from that of hot brew coffee is the higher fruity and floral flavor of cold brew coffee.³⁰ This was attributed to the detection of linalool, furfuryl acetate,

Table 1. Aroma-active compounds in cold brew coffees of different roasting degrees

Volatile compounds	Odor ^a	Threshold ($\mu\text{g L}^{-1}$) ^b	OAV ^c					
			Light roast	Medium-light roast	Medium roast	Medium-dark roast	Dark roast	Ultra-dark roast
1,6-Octadien-3-ol, 3,7-dimethyl-	Citrus, floral, sweet	0.006	–	1.14	1.63	–	–	–
Furfuryl acetate	Sweet, fruity	0.1	< 1	1.29	2.94	2	4.53	5.6
Benzeneacetaldehyde	Sweet, floral, cocoa	0.004	–	–	2.65	–	–	–
N-Methyl-2-pyrrolicarboxaldehyde	Roasted, nutty	0.037	< 1	1.08	1.66	1.54	2.77	2.45
2-Methoxy-4-vinylphenol	Sweet, clove, phenolic	0.003	3.55	5.28	5.89	6.86	8.43	4.01
3-Ethylphenol	Mouldy, leather, ink	0.00085	–	–	4.6	–	5.07	–
Guaiacol	Phenolic, smoke	0.007	< 1	1.53	3.64	2.47	6.88	10.98
2-Ethyl-6-methylpyrazine	Roasted	0.04	< 1	1.52	1.93	2.26	2.91	3.52
2-Ethyl-5-methylpyrazine	Bean, nutty	0.1	< 1	< 1	< 1	< 1	1.04	1.24
2,6-Diethylpyrazine	Nutty, hazelnut	0.006	< 1	2.46	2.84	3.46	4.21	6.54
Pyrazine, 3-ethyl-2,5-dimethyl-	Roasted, nutty	0.0086	2.61	7.08	6.84	10.88	13.54	20.03
Pyrazine, 2-ethyl-3,5-dimethyl-	Roasted, nutty	0.002	–	–	38.63	–	–	–

Note: A minus sign (–) means not detected because of the concentration of the given compound being below the detection limit.

^a Odor description found in the literature.^{3,10}

^b Odor thresholds in water taken from the literature.^{3,10}

^c OAV, odor activity value. Only OAV values > 1 are presented.

phenylethylaldehyde and 4-ethyl-2-methoxyphenol with sweet and fruity flavors in the present study. In terms of OAV, medium roast cold brew coffee has absolute advantages in fruit flavor and flower flavor. It is speculated that the medium roast cold brew coffee is outstanding in terms of nutty and roasted flavors, with a balance of fruity and floral flavors, which may be an important reason for its popularity among consumers.

As shown in Fig. 4(A), the W1C, W1S, W2S, W2W, W3C, W3S, W5C and W6S sensors had low responses in the cold brew coffee samples, and the signal intensities were very similar in these samples, making it difficult to distinguish the differences between them. However, the signal intensities of cold brew coffee with different roasting degrees showed significant differences on the W5S and W1W sensors. In particular, the response values on the W1W sensor showed a significant increasing trend with increasing roast degree, which is very favorable for indicating and measuring the roast degree. The W1W sensor is sensitive to sulfur-containing compounds, and the difference in response values may be an important indicator for differentiating coffee samples. Volatile sulfur-containing compounds in coffee may be related to the oxidation and decomposition of proteins during roasting, or may be produced by sulfur-containing amino acids (cystine, cysteine, methionine) via the Strecker degradation pathway.²⁵ However, the sensitivity of mass spectrometry detector to sulfur compounds is low, and a large number of such compounds cannot be found in the identification results of aroma compounds. The W5S sensor is very sensitive to these nitrogen/oxygen-containing products of the Maillard reaction. The volatile aroma compounds generated by Maillard reaction did not show a significant increasing trend during the baking process from light to ultra-dark, which made it difficult for W5S sensor to strictly distinguish the differences between different samples. As for the results of electronic tongue analysis (Fig. 4B), there is no significant difference in the richness of different coffee samples. However, with the increase of roasting degree, the intensity of acidity gradually decreases, whereas the bitterness and astringency are significantly enhanced. This is mainly because, with the increase of roasting degree, heterocyclic compounds with bitter and astringent taste produced by Maillard reaction gradually occupy the dominant position, making the bitter and astringent taste of coffee increase.³¹

Effect of freeze-drying on the flavor of cold brew coffee

The effect of freeze-drying on the flavor profile of medium roasted cold brew coffee, which has a strong flavor profile and is popular with consumers, is worthy of attention. Because of strong similarity of aroma composition and content between medium roast and medium-dark roast coffee (Fig. 2), freeze-drying was performed on both sample groups. Through OPLS-DA analysis of the aroma compounds of cold brew coffee, it is found that the aroma characteristics of coffee have changed significantly after freeze-drying, which is completely distinguished from the original coffee liquid (see Supporting information, Fig. S1). Moreover, the difference in the degree of roasting did not seem to have a significant impact. Furthermore, by comparison, it was found that freeze-drying mainly caused a reduction in the aroma compound content of cold brew coffee (Fig. 5). The loss of aroma compounds was more significant in the medium-dark roasted coffees. The same significant reduction in the content of aroma compounds was found when freeze-drying treatments were performed on blended pumpkin, mango and jujube juice.³² For medium roast cold brew coffee, the freeze-drying treatment mainly caused the

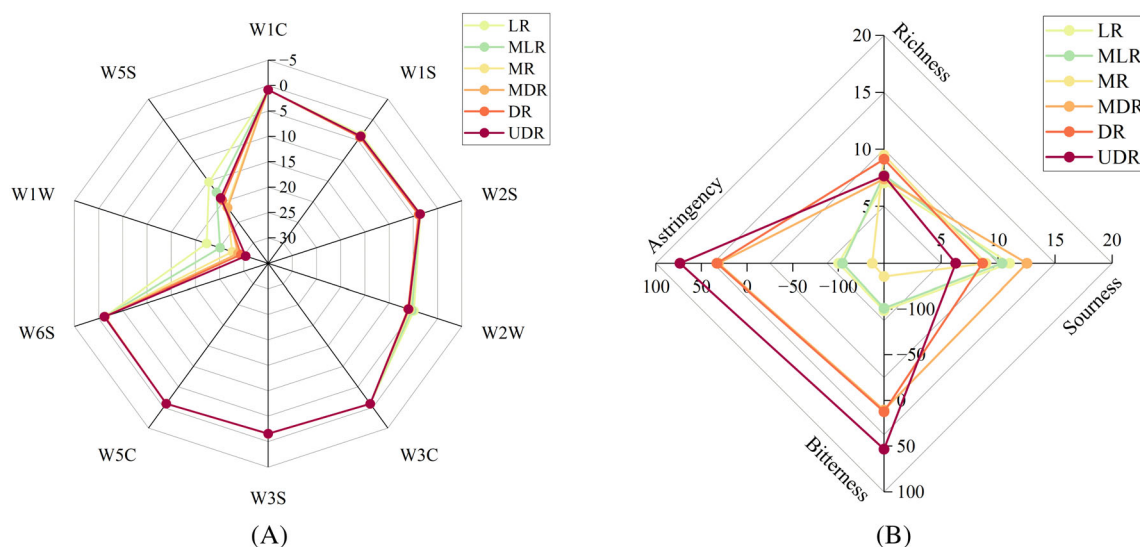


Figure 4. Radar plot of (A) electronic nose and (B) electronic tongue sensor response in cold brew coffees of different roasting degrees. LR, light roast; MLR, medium-light roast; MR, medium roast; MDR, medium-dark roast; DR, dark roast; UDR, ultra-dark roast.

loss of pyrazine and furan compounds. On this basis, the phenolic and ketone compounds in medium-dark roasted coffee also showed a significant decrease. Previous studies have shown that the degradation of amino acids or oxidation of unsaturated fatty acids will also occur during the freeze-drying process, producing aldehydes and ketones.³³ However, this phenomenon was not found in the present study, which may be a result of amino acid differences in the substrates. Generally speaking, high temperature is conducive to the occurrence of Maillard reaction and produces pyrazine compounds with roasted flavor. Gou *et al.*³⁴ found that, when multi-stage variable temperature freeze-drying was used, a variety of chemical and enzymatic reactions, including the Maillard reaction, occurred during the gradual increase in the temperature of jujuba slices samples from freezing temperature to heating plate temperature (above 60 °C), which additionally produced new aroma compounds such as esters and pyrazines to compensate for the aroma escape process. Because of the low temperature (below 40 °C) throughout the freeze-drying process in the present study, coupled with the low content of sugars and amino acids as substrates in coffee, favorable conditions for the occurrence of the Maillard reaction could not be created, and thus could only be reflected in the significant decrease in the content of aroma compounds caused by aroma volatilization, such as 3-ethyl-2,5-dimethylpyrazine and 2-ethyl-3,5-dimethylpyrazine.

The sharp decrease in the content of aroma compounds cannot directly determine the aroma quality of cold brew coffee, and the contribution of aroma compounds to the aroma characteristics of coffee needs to be further investigated using OAV.³² The comparison revealed that the decrease in the content of aroma compounds also directly affected the OAV of cold brew coffee, which in turn affected its aroma characteristics (Table 2). The total OAVs for medium-dark roasted cold brew coffee declined by more than 60% compared to a lower 52% decline for medium roasted cold brew coffee. A very similar phenomenon of aroma loss (56.7–63.5% reduction) was also found in an investigation of the effect of freeze-drying processing on the flavor profile of fruit juices as conducted by Yuan *et al.*³² However, freeze-drying still shows significant advantages in terms of aroma compound

retention compared to conventional thermal drying methods (e.g. hot air drying, solar drying and heat-pump drying).³⁵ The differences in retention of different aroma compounds depend on both the binding effect of the aroma compounds to the components of the coffee matrix, and the volatility of the aroma compounds themselves. In addition, a more important factor is the initial content of the relevant aroma compounds in the coffee liquid prior to the freeze-drying process. For example, 2-methoxy-4-vinylphenol and 3-ethyl-2,5-dimethylpyrazine were present in high levels in the coffee liquid, which results in relatively low losses after freeze-drying. The 2-ethyl-3,5-dimethylpyrazine with nutty flavor completely dissipated after freeze-drying. Only 2,6-diethylpyrazine and 3-ethyl-2,5-dimethylpyrazine retained after freeze-drying in medium roast cold brew coffees are still able to contribute nutty flavors. Furfuryl acetate and 4-vinyl-2-methoxyphenol, which have fruit and sweet taste,³ are still retained a lot in medium roasted coffee. Guaiacol with a smoky flavor will cause people to have a sense of disgust,¹⁰ which is significantly reduced after freeze-drying treatment, and the OAV value in medium roast coffee is lower, which has a positive effect on increasing consumer acceptance. In conclusion, the results of the analyses based on the OAV values revealed that, compared to medium-dark roasted cold brew coffee, medium roasted cold brew coffee after freeze-drying treatment are more dominant in the aroma profile qualities that have a positive contribution such as nutty, fruity and sweetness. This was also accompanied by a weaker smoky flavor, which is very beneficial for a high-quality drinking experience.

Before and after freeze-drying, the electronic nose analysis results of coffee samples are similar to the previous results (Fig. 6A). Only W1W and W5S sensors can distinguish the differences between different samples. The loss of aroma release caused by freeze-drying is also reflected in the measurement results of electronic nose, especially sulfur compounds and heterocyclic compounds. After freeze-drying, the electronic nose measurement results of medium and medium-dark roasted coffee samples are very consistent, and it is difficult to distinguish the difference between them. In addition to the aroma quality (more intense floral and fruity aroma) of cold brew coffee, its decreased

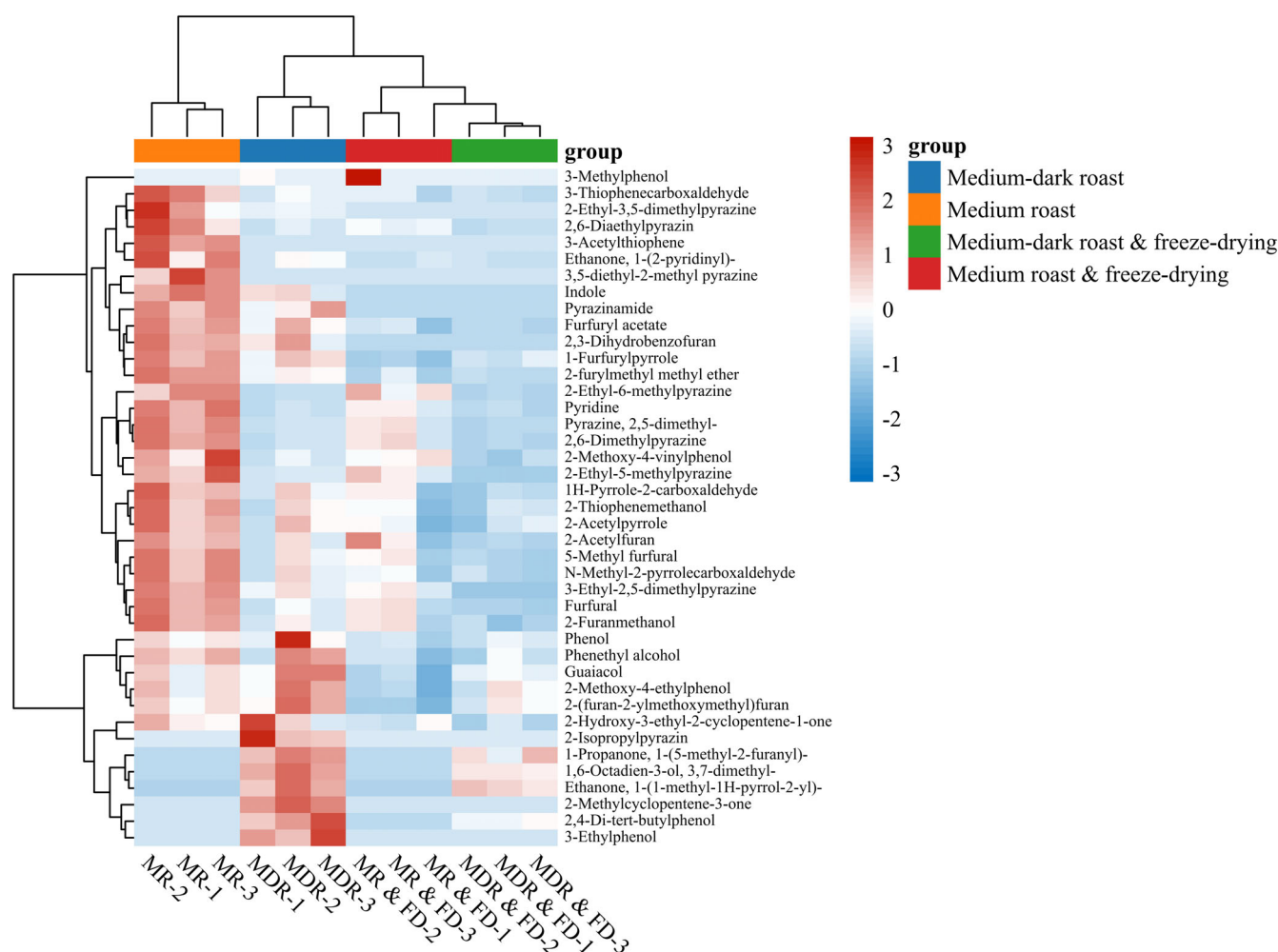


Figure 5. Heat map of the type and content of aroma compounds in cold brew coffee before and after freeze-drying. MR, medium roast; MDR, medium-dark roast; MR & FD, medium roast & freeze-drying; MDR & FD, medium-dark roast & freeze-drying.

Table 2. Aroma-active compounds in cold brew coffee before and after freeze drying

Volatile compounds	Odor ^a	Threshold ($\mu\text{g L}^{-1}$) ^b	OAV ^c			
			Medium roast	Medium roast & freeze-drying	Medium-dark roast	Medium-dark roast & freeze-drying
1,6-Octadien-3-ol, 3,7-dimethyl-	Citrus, floral, sweet	0.006	–	–	1.27	< 1
Furfuryl acetate	Sweet, fruity	0.1	3.41	1.46	2.44	1.39
3-Ethylphenol	Mouldy, leather, ink	0.00085	–	–	1.64	–
2-Methoxy- 4-vinylphenol	Sweet, clove, phenolic	0.003	4.31	2.51	1.61	< 1
Guaiacol	Phenolic, smoke	0.007	1.2	< 1	1.54	< 1
2,6-Diaethylpyrazin	Nutty, hazelnut	0.006	6.05	2.1	1.3	< 1
Pyrazine, 3-ethyl- 2,5-dimethyl-	Roasted, nutty	0.0086	7.57	3.61	3.54	–

Note: Only OAV values greater than 1 are presented. A minus sign (–) means not detected because of the concentration of the given compound being below the detection limit.

^a Odor description found in the literature.^{3,10}

^b Odor thresholds in water taken from the literature.^{3,10}

^c OAV, odor activity value.

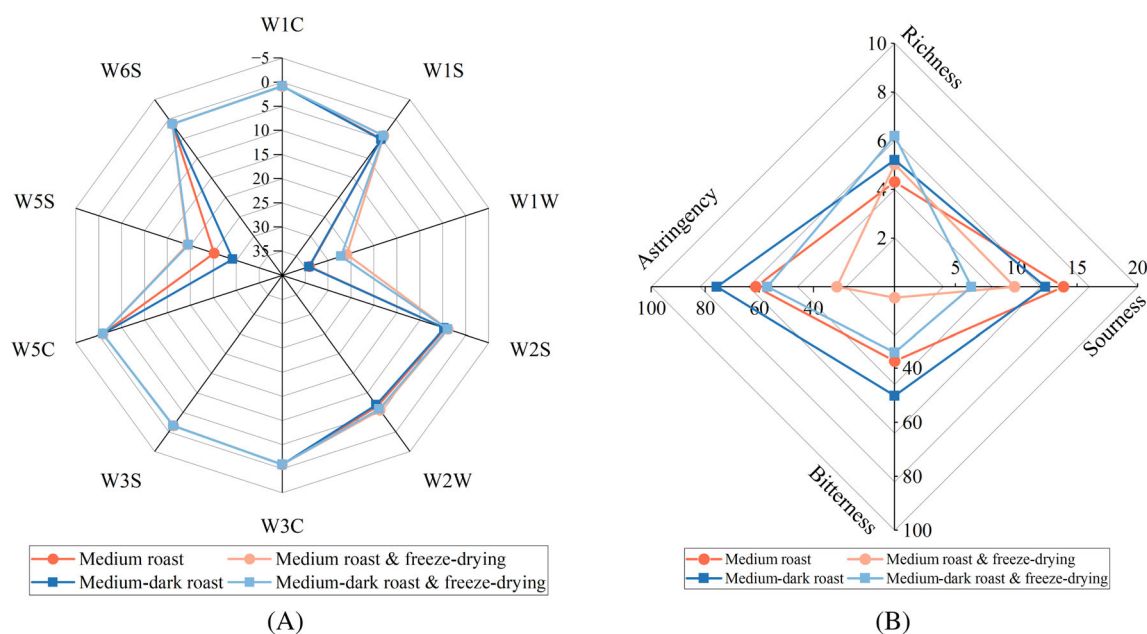


Figure 6. Radar plot of (A) electronic nose and (B) electronic tongue sensor response of cold brew coffee before and after freeze-drying.

bitterness and acidity are also important reasons for consumers to prefer it.³⁰ According to the results of the electronic tongue analysis (Fig. 6B), the taste richness of cold brew coffee has significantly improved after the freeze-drying treatment, which will bring a richer taste to consumers. Sourness, bitterness and astringency are tastes that consumers are unwilling to accept, which will reduce their preference for cold brew coffee.² From the experimental results, these three taste characteristics were significantly reduced after freeze-drying treatment, especially for the medium-roasted coffee. The sour and astringent taste of coffee mainly originates from non-volatile substances such as caffeine and quinic acid, but some aroma compounds also contribute to the bitter and astringent taste of coffee.³⁶ Although the freeze-drying process does not result in the loss of non-volatile content, the abrupt reduction of aroma compounds with bitterness and astringency will also reduce the bitterness and astringency of the coffee, especially in the medium-dark roasted coffee after freeze-drying, where there are almost no acid volatile compounds. In general, in terms of taste characteristics, after freeze-drying, the acidity, bitterness and astringency of cold brew coffee have significantly decreased, which is very positive for increasing consumers' drinking preference. Therefore, compared with medium-dark roasting, medium roast cold brew coffee is more suitable for freeze-drying processing, because of less loss of aroma compounds and lower bitterness, astringency and acidity intensity.

CONCLUSIONS

The present study has conducted a comprehensive comparative analysis of the aroma characteristics of cold brew coffee with roasting degrees ranging from light to ultra-dark and investigated the effect of freeze-drying on the flavor of cold brew coffee. Pyrazines and heterocyclic compounds were the main contributors and gradually dominated with the increasing degree of roasting. Pyridine might have potential to be served as a marker of the roast degree. The medium roasted cold brew coffee had the richest variety of aroma compounds, especially containing linalool, 2-ethyl-3,5-dimethylpyrazine,

phenylethyl aldehyde and 3-thiophenylaldehyde, which were significantly different from other coffee samples. The fruit and floral flavor of cold brew coffee were critical to differentiate them from hot brew. The OAV analysis revealed that pyrazines, linalool, furfuryl acetate, phenylethylaldehyde and 4-vinyl-2-methoxyphenol were the main contributors to the roasted, floral and fruity aromas of coffee. The stronger floral and fruity aroma in the medium roast cold brew coffee may be related to its the higher consumer acceptance and enjoyment. Medium roasted cold brew coffee after freeze-drying treatment was more dominant in the nutty, fruity and sweetness. The present study found that medium-roasted cold brew coffee is more suitable for freeze-drying processing. The elucidation of the material basis of aroma characteristics of cold-brewed coffee with different roasting degrees is of great value in guiding industrial production. Application of freeze-drying for cold brew coffee will also promote the convenience of cold brew coffee drinking. However, future studies are needed to work on minimizing favorable aroma loss during the freeze-drying process.

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CONFLICTS OF INTEREST

The authors declare that they have no conflicts of interest.

DATA AVAILABILITY STATEMENT

Research data are not shared.

SUPPORTING INFORMATION

Supporting information may be found in the online version of this article.

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